



Cambridge IGCSE™

BIOLOGY

0610/61

Paper 6 Alternative to Practical

October/November 2022

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2022 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

This document consists of **9** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	<u>'List rule' guidance</u> For questions that require <i>n</i> responses (e.g. State two reasons ...): • The response should be read as continuous prose, even when numbered answer spaces are provided. • Any response marked <i>ignore</i> in the mark scheme should not count towards <i>n</i>. • Incorrect responses should not be awarded credit but will still count towards <i>n</i>. • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should not be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. • Non-contradictory responses after the first <i>n</i> responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Mark scheme abbreviations

- ; separates marking points
- / alternative responses for the same marking point
- R reject the response
- A accept the response
- I ignore the response
- ecf error carried forward
- AVP any valid point
- ora or reverse argument
- AW alternative wording
- underline actual word given must be used by candidate (grammatical variants excepted)
- () the word / phrase in brackets is not required but sets the context

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Question	Answer	Marks	Guidance
1(a)(i)	table drawn with minimum of two , columns / rows and with a line separating headings from the body of the table ; appropriate headings for each, column / row ; nine correct results recorded ; three correct averages ;	4	
1(a)(ii)	as an indicator / so that the iodine would change colour / to see the change / to react with the iodine / to give a positive result with iodine / AW ;	1	
1(a)(iii)	(health drink / solution) B <i>because B</i> needs more (drops / greater volume of) iodine (solution to stay blue-black) ;	1	
1(a)(iv)	the, number / amount, of drops (of iodine solution added) ;	1	
1(a)(v)	to prevent contamination / so the solutions are not mixed / AW ;	1	

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Question	Answer	Marks	Guidance
1(a)(vi)	<i>error:</i> volume (of iodine) not directly measured / inconsistent drop size / using a (dropping) pipette ; <i>improvement:</i> use a syringe / use a burette / use a graduated pipette / use a measuring cylinder / measure the (iodine) volume ; or <i>error:</i> judgement of the endpoint / endpoint is subjective / endpoint is qualitative ; <i>improvement:</i> compare to a colour standard / compare to a colour chart / compare to one that had already changed colour / colorimeter ; or <i>error:</i> different degree of , shaking / mixing / stirring ; <i>improvement:</i> shake at a fixed speed / use a (mechanical) stirrer / AW ; or <i>error:</i> different time of , mixing / shaking ; <i>improvement:</i> shake / stir , for a fixed time ;	2	error must match improvement

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Question	Answer	Marks	Guidance
1(b)(i)	a result that does not fit the , pattern / trend (of the other results) or was, very different / much higher, than the other two, results / trials 1 and 2 or was outside the range of the other results / is an outlier / AW ;	1	
1(b)(ii)	removed it / did not include it / ignored it / excluded it / only used the other two results / AW ;	1	
1(b)(iii)	axes labelled with units that match the table headers ; linear scale and plotting area that occupies at least half the grid in both directions ; six points plotted accurately $\pm$ half a small square ; suitable line drawn ;	4	
1(b)(iv)	indication on the graph of where reading taken ; correct value from candidate's graph at seven drops ($\pm \frac{1}{2}$ a small square) ;	2	
1(b)(v)	(Benedict's reagent / it) must be heated ;	1	

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Question	Answer	Marks	Guidance
2(a)(i)	line EF = 70 ± 1 (mm) ; 230 ;;	3	MP1 correct measurement of line EF MP2 correct answer to any number of sig figs ($70 \div 0.3 =$) 233.33 MP3 correct rounding of their calculation to two significant figures A 240 if measurement is 71 mm (from 236.7)
2(a)(ii)	outline: single, clear unbroken line ; size: drawing must be at least 70 mm long ; detail 1: 8-10 segments drawn ; detail 2: black line thickness thicker near the top ;	4	
2(b)	<i>any two from:</i> stripes / stripy pattern (Monarch) or one stripe (QA) ; antennae (Monarch) or no antennae (QA) ; rounded (pro)legs (Monarch) or spiky / pointy legs (QA) ; (visible) head (Monarch) or no (visible) head (QA) ; spikes on two segments / no spikes / smooth 'skin' (Monarch) or many spikes / spikes on each segment / AW (QA) ; QA , dark / much darker / mostly black ;	2	
2(c)(i)	<i>any two from:</i> species (of butterfly / caterpillar) ; age ; mass / amount, of food ; number / amount, of caterpillars ; all kept in glasshouses ; number / amount, of days ;	2	A time intervals / time of recording
2(c)(ii)	(–)3.8 (%) ;;;	3	MP1 selection of data (see 25 and 26 somewhere) MP2 correct calculation ($(26-25) \div 26 \times 100 = 3.846$) to any number of decimal places MP3 correct rounding of their calculation to one decimal place

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Question	Answer	Marks	Guidance
2(c)(iii)	25 (°C) ;	1	
3	<i>six from:</i> 1 <i>independent variable</i> at least two different temperatures , tested / stated ; 2 <i>dependent variable</i> counting bubbles / measuring volume of gas given off ; 3, 4, 5 <i>variables kept constant - max three from ;;;</i> set time period (for measurement) species / type of, plant / plant tissue size / mass of plant or number of leaves age of plant carbon dioxide / HCO_3^- concentration light intensity / same lamp / same light bulb / AW pH of water nutrient content of water volume of water 6, 7 <i>method – max two from ;;</i> use of gas syringe / upturned measuring cylinder or test-tube method of controlling the temperature fixed distance from light source use of heat shield equilibration time use of oxygen probe AVP 8 repeat investigation (at least) twice / at least two replicates ;	6	A colour change of hydrogencarbonate indicator / measure, oxygen or carbon dioxide, concentration